

Tests of numerical errors between Matlab and Python functions

This document is a report of all the numerical errors that were measured between Numpy/Scipy and Matlab's functions for sound processing .

First : Interpolations :

The Outer Middle Model contains two steps of interpolations , the two interpolated lists are the added in tfOuterMiddle, here is the error resulting of the same computation between Matlab and Numpy's interpolation.

1 : Outer Middle Model :

Here are the measures done on the list called OuterMiddle of the data_glasberg matlab file reproduced in python.

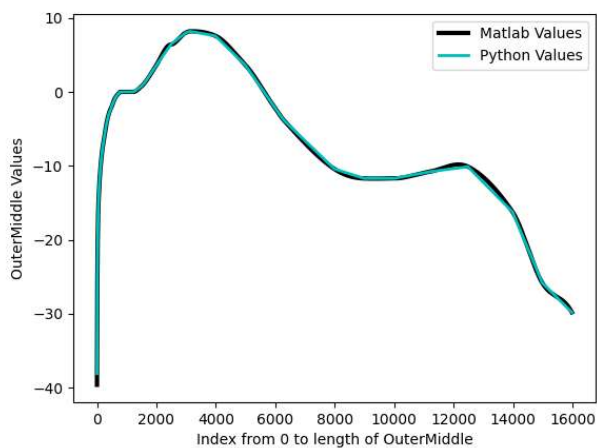
Maximum error for outerMiddle is : 1.5200000000000031

* Maximum error is $\max(\text{abs}(\text{error}))$

Average error for outerMiddle is 0.1826141705261848

Quadratic error for outerMiddle is : 0.06420167375087449

Percentage of error : $\text{ratio} = (\text{max_error} / \text{average_MatlabOuterMid})$ is : 3.1797143223154336 %



FILTERS :

Fir2 filter :

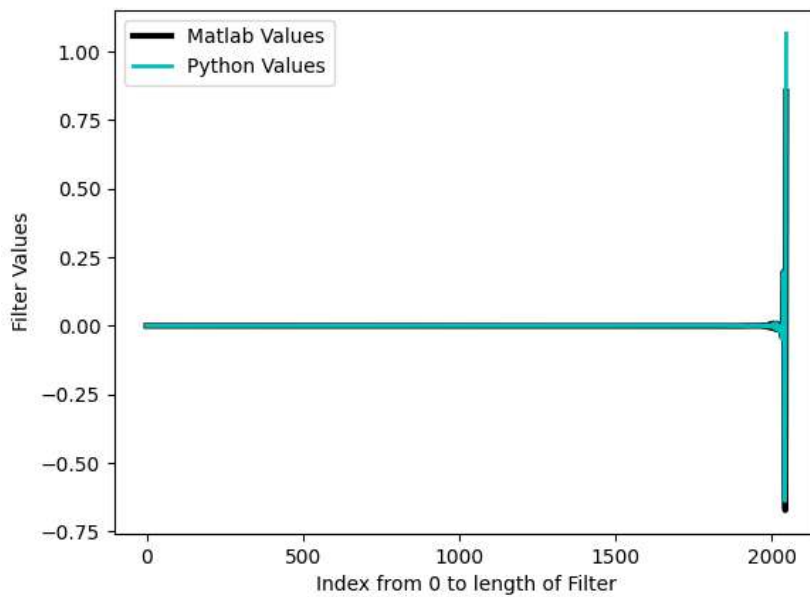
The parameters are 1997 model with Free model .

Maximum error for Filter is : 0.3200670631705158

Average error for Filter is 0.0009037093750747852

Quadratic error for Filter is : 0.00014586153447515663

Percentage of error : ratio = (max_error / (max - min)) is : 18.421703285903938 %



Enhanced : The problem was that The python version did not have the right sampling,

Fir2 filter 2 :

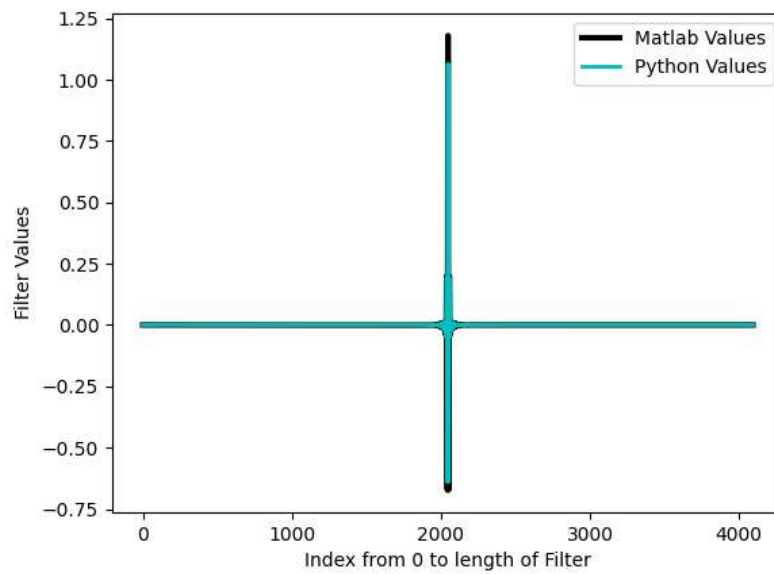
Maximum error for Filter is : 0.32527708982965475

Average error for Filter is 0.0008921720402584933

Quadratic error for Filter is : 0.00014654680690834955

Percentage of error : (max_error / average_MatlabOuterMiddle) is : 17.563824618716307 %

Quadratic Percentage of error : (Quad_error / range of values) is : 0.007913014766330696 %



FILTFILT :

Maximum error for filtfilt is : 0.00045844424666551973

Average error for filtfilt is 6.884825540168854e-07

Quadratic error for filtfilt is : 3.751371605876241e-11

Percentage of error : (max_error / average_MatlabOuterMiddle) is : 15.173997665704443 %

Quadratic Percentage of error : ratio = (Quad_error / range of values) is :

1.2416625228647962e-06 %

